

Classroom speaking script

Digital Image Processing in Robotics: From Camera Pixels to Robot Action

MuJoCo Vision-Guided Multi-Object Pick-and-Place | B.Tech | 4 October 2026

Use one 30-60 second explanation per slide. The editable PowerPoint contains the same explanations plus three likely professor questions per slide.

01 Digital Image Processing in Robotics

This project asks how a robot can turn camera pixels into a physical action. In this MuJoCo demonstration, I choose a box color, inspect the image-processing stages, select an instance when two boxes share the color, and then let a Panda arm pick it into a tray. The target position and angle come from the rendered camera image. Simulator object coordinates are used later only to measure errors. The Panda robot model and workshop DLS components are reused with attribution. The camera calibration, vision pipeline, controls, tests and integration are the new project work. The results shown here come from completed simulation runs.

02 From camera pixels to robot action

The pipeline has two connected parts. First, digital image processing prepares the camera image: optional simulated noise, median filtering, HSV color thresholding and morphological cleanup. Next, perception interprets the cleaned pixels as a rectangular object, estimates its centre and long-edge angle, and maps these to world coordinates. A validity and reachability gate must pass before motion begins. Inverse kinematics finds joint values for the desired hand poses, and the position servos execute those joint commands. Planning chooses the approach and transfer sequence. The dashboard pauses between these stages so each result can be explained before the robot moves.

03 A calibrated scene with selectable targets

The scene contains a Panda arm, a flat table, a tray and four elongated boxes: red, green and two blue instances. The fixed overhead camera renders a 640 by 480 RGB frame. The observation pose keeps the boxes visible before capture. One matching color region is selected automatically, while two blue regions require a click in the camera preview. Box setup uses metre and degree controls within a conservative workspace. Each requested placement also needs joint-limited IK and sampled table-clearance checks. The validated area is smaller than the robot's theoretical workspace, and the demonstration does not claim universal collision-free reachability.

04 Median filtering reduces simulated noise

MuJoCo's RGB image is usually clean, so the noise shown here is deliberately simulated. A fixed random seed changes selected pixels to black or white, creating salt-and-pepper noise. The displayed example corrupts eight percent of pixels. A three by three median filter sorts each channel's local values and chooses the middle value, so an isolated extreme value usually does not dominate the result. The filter runs before segmentation. It greatly reduced the measured image difference in this example, but can soften small details and does not remove every kind of noise. The other boxes are distractor objects, not image noise.

05 HSV segmentation and mask cleanup

The filtered RGB image is converted to HSV: hue describes color, saturation describes its strength, and value describes brightness. Threshold ranges produce a binary mask. Matching pixels are white and everything else is black. Red uses two hue ranges because hue wraps around the scale. A three by three opening applies erosion followed by dilation to remove small isolated foreground regions. Closing reverses that order to fill small gaps. The median filter already removes most impulse specks in these rendered frames, so the extra change may be small. Morphology improves the mask but does not by itself decide which region is a valid box.

06 Detect a rectangle and estimate its angle

Contours group the white pixels into candidate regions. The program rejects regions with implausible area, side length, aspect ratio or rectangular fill. OpenCV `minAreaRect` fits a rotated rectangle to each accepted outline. Its centre is the image location, and the longest edge supplies an orientation cue. The fill ratio is a shape-validity measure, not a probability of correctness. An elongated box has a clear long edge; a square would have ambiguous yaw. The image angle is not sent directly to the robot. Both edge endpoints are converted through camera calibration to obtain a world direction, and world yaw is normalized modulo 180 degrees.

07 Pixels become world position and yaw

A pixel is a viewing direction, not a distance in metres. The camera's field of view and image size give the focal length, while its position and rotation define the ray in world coordinates. The program intersects that ray with the known box-top plane at z equal to 0.27 metres. It then subtracts half the box height to report the box centre. Using the tabletop plane would give the wrong geometry. Mapping the longest-edge endpoints gives the world yaw. Calibration matters because any camera or height error changes the target estimate. These results assume a fixed overhead camera, a flat table and uniform box height.

08 Vision supplies a pose; IK and control move the arm

The selected image estimate defines a box centre and yaw. The planner builds hand waypoints above the box, down to grasp height, up for lifting, over the tray and down to release. Damped least-squares IK uses a Jacobian to find bounded joint targets for each pose. All waypoints are checked before starting. The arm's position servos then follow smoothly interpolated joint commands, while the gripper opens and closes through its tendon actuator. The actual box is transported by finger contact, friction and gravity. There is no box teleport or object weld. A final retreat and return clear the camera for the next demonstration.

09 The same perception-to-action idea in industry

The general idea appears in industrial robot systems: a camera measures the scene, perception identifies useful objects or workpiece locations, and a robot acts on those estimates. ABB documents vision-guided workpiece handling, picking and packing, and warehouse item picking as real examples. These applications should be separated from this classroom implementation. Commercial systems may use three-dimensional sensors, learned perception, suction tools, moving conveyors and more advanced planning. Our project uses simple HSV ranges, a known object plane, rectangular boxes and a simulated two-finger gripper. It demonstrates the reasoning chain clearly, rather than reproducing an industrial product's capabilities.

10 Measured results and practical limits

The completed test set used ten physical pick-and-place runs across blue, red and green targets, several positions and yaw angles, and zero, three and eight percent simulated noise. It also tested the second blue instance. Every selected object lifted, was released inside the tray and left the fingers open. Three negative checks correctly refused a missing target, an outside-workspace target and an unreachable IK pose. Position and angle errors are calculated only after image detection. The initial failed blue-threshold run is preserved, and these reported numbers come from the corrected run. Limits include fixed heights, controlled lighting, occlusion and sampled rather than exhaustive path checking.

11 Live Demo: Vision-Guided Pick and Place

Now that we have seen the camera-to-action pipeline and measured results, let us watch the completed MuJoCo simulation. Click the embedded video on slide 11. This recording shows the completed simulation. As it plays, point to the camera and image-processing views, then follow the robot through approach, grasp, lift, transfer and release.

12 Thank you

Thank you. Feel free to ask questions. Keep this slide on screen for discussion and return to the pipeline or results slides when needed.

Live demo explanation

Backup for the live demo

If the live MuJoCo window fails or runs too slowly, go to slide 11 and play the embedded recording. Tell the class: "This recording shows the completed simulation; it is not live." Continue explaining the same camera-to-action steps while the recording plays.

1. Reset and choose BLUE. Point to both blue boxes among the red and green distractors. Turn simulated noise on and keep Step-through enabled.
2. Click Detect / start stages, then Next stage. Explain: "This is a real RGB frame rendered by MuJoCo. The black and white specks are optional simulated salt-and-pepper noise, not the other boxes."
3. At the filtered view, say: "The 3 x 3 median chooses a neighborhood middle value, reducing isolated extreme pixels. It does not remove every kind of noise."
4. At the mask, say: "HSV thresholds make matching blue pixels white. Opening removes isolated regions; closing fills small gaps. Segmentation creates the mask; detection validates rectangular regions."
5. Click the specific blue box in the RGB preview. Explain the fitted rectangle, centre point and longest edge. Show that changing color changes the mask while the distractor boxes remain in the RGB image.
6. Continue to localization. Say: "Camera calibration maps a pixel ray to the known box-top plane. The result is a position in metres and a world yaw angle, normalized modulo 180 degrees."
7. At Reachability checked, point out that the robot has not moved. Explain: "IK checks the requested hand poses under joint limits before the controller receives commands."
8. Click Next stage to pick. Name the visible approach, descent, grasp, lift, transfer and release. Use Pause / resume robot if needed. Say: "The box moves through finger contact physics, not a teleport or attachment weld."
9. Reset, select absent YELLOW and repeat until detection. Point to NOT FOUND and REJECTED. Explain that the robot refuses before movement.
10. For repositioning, use metres for X/Y and degrees for yaw. Validate and place checks workspace bounds, box separation, IK and sampled table clearance. Reset before another pick.

Measured evidence

Completed set: 10/10 physical picks; 3/3 correct refusals; mean position error 0.200 cm (max 0.330 cm); mean yaw error 0.296 degrees (max 0.964 degrees). Ground truth is used only after detection for evaluation. These finite results do not establish reliability for arbitrary scenes.

Attribution and limits

Panda model: reused from the supplied Solutions repository / MuJoCo Menagerie, Apache 2.0. Workshop DLS utilities: Debojit Das, MIT. New work: vision, calibration, controls, integration and validation. Assumptions: fixed camera, flat table, uniform 4 cm box height and controlled colors. Limits: occlusion, lighting changes, no variable-height inference and sampled rather than exhaustive collision checking.

Industry examples are separate from this implementation. Sources: ABB Vision Systems (new.abb.com/products/robotics/application-equipment-and-accessories/vision-systems) and ABB Robotic Item Picker (abb.com/global/en/areas/robotics/solutions/functional-modules/item-picking-family/robotic-item-picker).